FSDL
ASSIGNMENT-04

AIM:-
Write server side script in PHP to perform form validation and create database application using PHP and MySQL to perform insert/update/delete/search operations.

OBJECTIVES:-
1) To understand server-side scripting.
2) To learn database connectivity using PHP-MySQL.
3) To perform insert, update, delete and search operations on database.

THEORY:-
Q) PHP ARCHITECTURE:
PHP (Hypertext Preprocessor) follows a server-side scripting architecture. It works as follows:
- The client (browser) sends a request to the server.
- The web-server (like Apache/XAMPP) forwards the request to PHP engine.
- The PHP engine processes the script (logic, database operations etc).
- If needed, PHP communicates with MySQL database.
- The server returns the generated HTML response to the browser.

* Key Components:
- Client → Browser (Chrome, Edge)
- Web server → Apache/Nginx
- PHP interpretor → Executes PHP code
- Database → MySQL

PHP is embedded inside HTML and executed on the server, not on the client.

Q) Steps for Database connectivity in PHP.
→ To connect PHP with MySQL, follow these steps:

1) Create Database:
CREATE DATABASE student_db;

2) Create Table:
CREATE TABLE students (
name VARCHAR(50);
id INT AUTO_INCREMENT PRIMARY KEY);

3) Connect PHP to MySQL:-
Using MySQLi:
```
$conn = mysqli_connect ("localhost", "root", " ", "student_db");
if (!$conn) {
    die ("connection failed:". mysqli_connect_error ());
```

4) Perform Operations:
Insert / Update / Delete / Select

5) Close Connection:
mysqli_close($conn);

FAQs:-
Q) What are advantages of server-side scripting?
→ · Better Security
· Database Interaction
· Dynamic Content
· Centralized Control
· Authentication Support.

What is XAMPP and phpMyAdmin?

1) XAMPP:
- A local server package
- Includes:
  - Apache (web server), MySQL (database), PHP (scripting language)
- Used to run PHP projects locally

2) phpMyAdmin:
- A web-based tool to manage MySQL databases
- Features:
  - Create/delete databases
  - Run SQL queries
  - Manage tables and records
- Comes bundled with XAMPP.

Q) What are two ways to connect a database in PHP?

→ 1) MySQLi :

```
$conn = mysqli_connect("localhost", "root", " ", "db_name");
```

2) PDO (PHP Data Objects):

```
$conn = new PDO("mysql:host=localhost;dbname=db_name", "root", "");
```

CONCLUSION:-

Thus we learnt server-side scripting using PHP and MySQL enables secure, dynamic and efficient database-driven web applications.

ASSIGNMENT 4

localhost/Assignment5/index.php

Student System | Library System

Student Registration

First Name

Last Name

Roll No / ID

Password

Contact Number

Register Student

Registered Students

Name	Roll No	Contact	Actions
Siddhi Karhekar	06	3273214	Edit Delete

XAMPP Control Panel v3.3.0 [Compiled: Apr 6th 2021]

XAMPP Control Panel v3.3.0

Modules

Service	Module	PID(s)	Port(s)	Actions				
✕	Apache	5900 616	80, 443	Stop	Admin	Config	Logs	
✕	MySQL	8808	3306	Stop	Admin	Config	Logs	
✕	FileZilla			Start	Admin	Config	Logs	
	Mercury			Start	Admin	Config	Logs	
✕	Tomcat			Start	Admin	Config	Logs	

Config
Netstat
Shell
Explorer
Services
Help
Quit

```
16:04:07 [main]   All prerequisites found
16:04:07 [main]   Initializing Modules
16:04:07 [main]   Starting Check-Timer
16:04:07 [main]   Control Panel Ready
16:04:09 [Apache] Attempting to start Apache app...
16:04:09 [Apache] Status change detected: running
16:04:10 [mysql]  Attempting to start MySQL app...
16:04:11 [mysql]  Status change detected: running
```